

PRATIK KAMATH

+61 439 839 218 ✦ Visa: Temporary Graduate Visa (Subclass 485)

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Education

Master of Information Technology – Major in Artificial Intelligence

June 2025

University of New South Wales

Sydney, NSW

- Recipient of the **International Student Award** for **academic merit** and commitment to studies.
- Distinction WAM** in core courses including Advanced Machine Learning and Data Structures & Algorithms.

Bachelor of Computer Engineering

May 2022

NMIMS University

Mumbai, India

- Published research on **financial time series forecasting using neural networks**, a comparative study [\[IEEE\]](#)

Technical Skills

Languages: Python, Java, C, JavaScript, HTML/CSS, SQL

Frameworks & Libraries: FastAPI, LangChain, PyTorch, scikit-learn, React, Next.js

Databases & Vector Stores: PostgreSQL, MongoDB, Redis, Pinecone

AI/ML & GenAI: LLM integration, RAG pipelines, Prompt Engineering, Embeddings, Model Evaluation

Cloud & DevOps: AWS (EC2, S3), Google Cloud Platform, Docker, Kubernetes, Linux, CI/CD (Jenkins)

Developer Tools & Practices: Git, REST APIs, Async Programming, Unit Testing (JUnit), Scalable API Design

Experience

Mark Computers

Oct 2025 – Dec 2025

Software Engineering Intern

- Engineered **enterprise-grade** automation across POS, accounting, HR, and inventory as measured by digitisation of multi-station petrol pump operations, using **Python (FastAPI) + PostgreSQL** with **Bootstrap** and **jQuery**.
- Reduced manual logging errors by **93%** by integrating **Multi-Product Dispenser (MPD)** hardware to capture real-time fuel sales and shift-wise meter readings.

L&T Realty

May 2021 – Jul 2021

Software Engineering Intern

- Delivered real-time price predictions for 5,000+ property listings as measured by handling 1,000+ daily inference requests, by designing an application using **Python**.
- Optimized manual analysis time by **73%** by productionizing a model serving pipeline using **Docker**, **Pydantic**, and **PostgreSQL**-backed feature queries.

Projects

AI-Powered Newsletter Generator with Event-Driven Architecture [\[GitHub\]](#) | Gemini, Inngest, FastAPI 2025

- Architected event-driven newsletter generation system using Google Gemini 2.5 Flash and Inngest workflow orchestration, enabling autonomous multi-step content creation from web research to publication.
- Boosted **concurrent task processing capacity by 40%** through asynchronous background job execution with Inngest, allowing parallel generation of newsletters across multiple topics.
- Integrated BrightData SERP API for real-time web intelligence, combining AI-powered content synthesis with live news aggregation to deliver contextually relevant newsletters.

RAG-Based Document Intelligence System [\[GitHub\]](#) | Google Gemini, Pinecone, Streamlit 2025

- Engineered retrieval-augmented generation (RAG) pipeline using Google Gemini and Pinecone vector database, achieving **up to 65% improvement in answer accuracy** compared to standalone LLM performance.
- Enhanced **retrieval precision by 30-40%** through metadata-enriched chunking strategy, generating contextual summaries and structural annotations for each document segment to improve similarity matching.

CeleStE Conference Management Platform [\[GitHub\]](#) | Python, React, React Native, Node.js, Docker, MongoDB 2025

- Spearheaded 6-member agile team as Product Owner to deliver full-stack conference platform serving **500+ attendees** with cross-platform web and mobile applications supporting real-time session booking and live engagement features.
- Designed scalable **Node.js** backend with microservices-inspired architecture, integrating **MongoDB** for document storage, **Redis** for caching and analytics, and **JWT authentication**, achieving **sub-200ms API response times**.
- Implemented real-time communication infrastructure with **WebSocket**-based chat, Q&A forums, and push notifications via Expo, while orchestrating Docker containerization and CI/CD deployment workflows.